

Chapter 18: Linear Independence

Example: We showed that the set

$$W = \left\{ \begin{bmatrix} r \\ s \\ r+2s \end{bmatrix} \mid r, s \in \mathbb{R} \right\}$$

forms a subspace, and that $S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \right\}$ is a spanning set for W .

$T = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} \right\}$ also spans W . Why is S “better”?

Because S is a linearly independent set.

A subspace U has many different sets whose span is U . How do we determine which sets are the most “efficient”?

Definition: The vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ in a vector space V are said to be **linearly independent**, if whenever $a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n = \mathbf{0}$,

$$a_1 = a_2 = \dots = a_n = 0.$$

The **trivial linear combination** of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is the one with every coefficient equal to zero:

Definition: The vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ in a vector space V are said to be **linearly dependent**, if there exists $a_1, a_2, \dots, a_n$ real numbers, such that

$$a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_n \mathbf{v}_n = \mathbf{0},$$

and not all $a_i = 0$

Example: Determine whether the vectors from $\mathbb{R}^3$

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} -1 \\ 4 \\ -2 \end{bmatrix}$$

are linearly independent.

$$a_1 \vec{v}_1 + a_2 \vec{v}_2 + a_3 \vec{v}_3 = \vec{0}$$

$$\Rightarrow \begin{cases} 3a_1 + a_2 - a_3 = 0 \\ 2a_1 + 3a_2 + 4a_3 = 0 \\ 2a_1 - 2a_3 = 0 \end{cases} \Rightarrow a_1 = a_2 = a_3 = 0$$

So $\vec{v}_1, \vec{v}_2, \vec{v}_3$ are linearly indep.

Note that $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ in $\mathbb{R}^n$ is a linearly independent set precisely when $A\mathbf{x} = \mathbf{0}$ has **only** the zero solution.

Independence Test

To verify if a set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is (linearly) independent, proceed as follows:

1. Set a linear combination equal to the zero vector: $a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_k \mathbf{v}_k = \mathbf{0}$
2. Solve the resulting linear homogeneous system.
3. If the only solution is $a_1 = a_2 = \dots = a_k = 0$, then the set is independent. Otherwise, the vectors are linearly dependent.

We can characterize linear dependence in another way:

Lemma: A set $D = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ of vectors in a vector space V is linearly dependent if and only if some vector in D is a linear combination of the others.

Proof: " $\Rightarrow$ " If D is L.D., then $\exists a_i \neq 0$, s.t.

$$a_1 \vec{v}_1 + \dots + a_n \vec{v}_n = \vec{0} \Rightarrow a_i = -\frac{a_1}{a_i} \vec{v}_1 - \dots - \frac{a_n}{a_i} \vec{v}_n.$$

So $\vec{v}_i$ is a linear comb. of the others.

" $\Leftarrow$ " If $\vec{v}_i$ is a linear comb. of the others, then

Example: Determine whether the vectors from M_{22} $\vec{v}_i = a_1 \vec{v}_1 + \dots + a_n \vec{v}_n$.

$$\mathbf{v}_1 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 0 & -3 \\ -1 & -1 \end{bmatrix} \Rightarrow a_1 \vec{v}_1 + \dots - \vec{v}_i + \dots + a_n \vec{v}_n = \vec{0}$$

are linearly independent.

$$a_1 \vec{v}_1 + a_2 \vec{v}_2 + a_3 \vec{v}_3 = \vec{0}$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 2 & -2 & -3 & 0 \\ 0 & 0 & -1 & 0 \\ 1 & 0 & -1 & 0 \end{array} \right] \Rightarrow a_1 = a_2 = a_3 = 0$$

$\Rightarrow$ they are L.D.

so $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are L.I.

Example: Determine whether the vectors from P_2

$$\mathbf{v}_1 = x^2 + x + 2, \mathbf{v}_2 = 2x^2 + 3x, \mathbf{v}_3 = 3x^2 + 2x + 2, \mathbf{v}_4 = x + 4$$

are linearly independent.

$$\begin{cases} a_1 + 2a_2 + 3a_3 = 0 \\ a_1 + 3a_2 + 2a_3 + a_4 = 0 \\ 2a_1 + 2a_3 + 4a_4 = 0 \end{cases} \Rightarrow r \leq 3 < 4, \text{ so inf. many solns, so } \mathbf{v}_1, \dots, \mathbf{v}_4 \text{ are L.D.}$$

Question: Is the set $S = \{\mathbf{0}\}$ linearly independent?

NO, $a_1 \cdot \vec{0} = \vec{0}$, $a_1 \neq 0$.

Corollary: If a set S of vectors contains $\mathbf{0}$, then S is **linearly dependent**.

Question: Is a set consisting of a single nonzero vector linearly dependent?

NO, $a_1 \cdot \vec{v}_1 = \vec{0}$, if $\vec{v}_1 \neq \vec{0}$ then $a_1 = 0$.

Question: When will a set of two distinct vectors $\{\mathbf{u}, \mathbf{v}\}$ be linearly independent?

If $\vec{u}, \vec{v}$ are L.D., $\exists a_1, a_2$ not all zero, s.t.
 $a_1 \cdot \vec{u} + a_2 \cdot \vec{v} = \vec{0}$.

assume $a_1 \neq 0$, then

$$\vec{u} = -\frac{a_2}{a_1} \cdot \vec{v}$$

so $\{\vec{u}, \vec{v}\}$ is L.D. iff one vector is a rescaling of the other one. (or parallel)

If $\langle \vec{u}, \vec{v} \rangle$ is L.I., then $\vec{u}, \vec{v}$ are NOT parallel to each other.